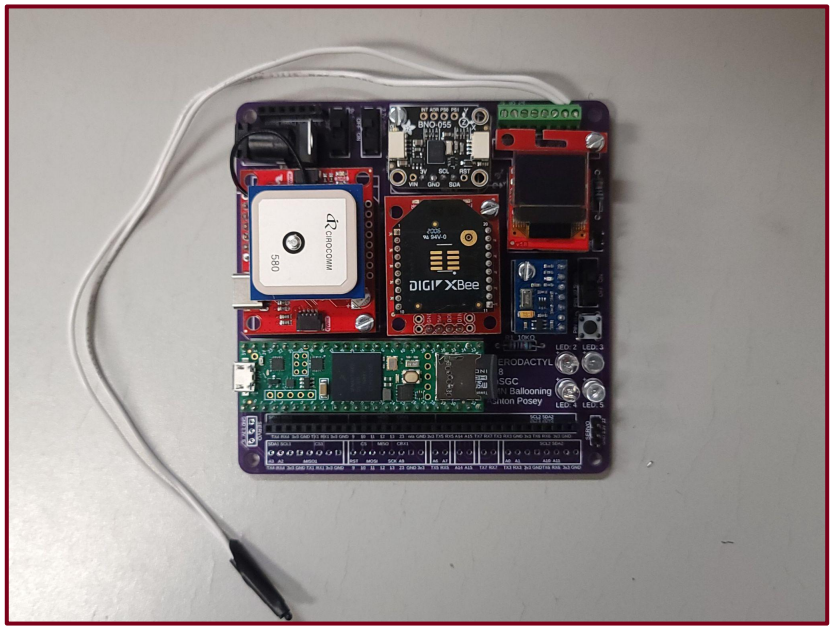
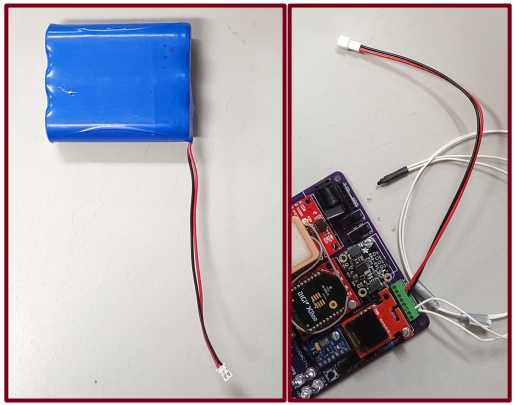
PTERODACTYL v3.8 Setup

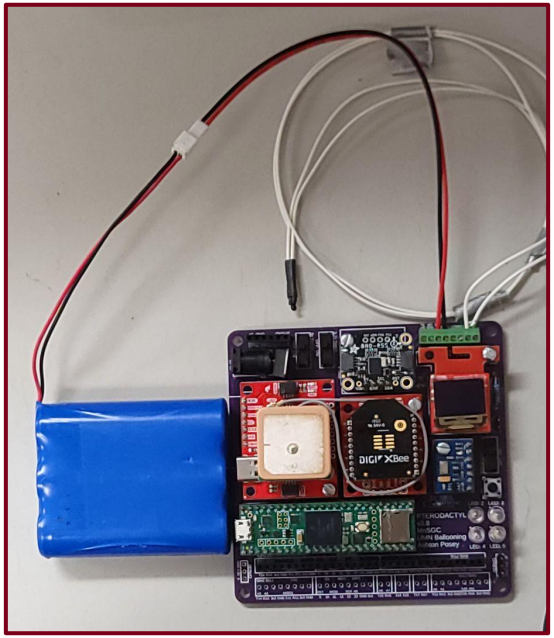


MN Space Grant, summer 2024: contact Sean Gilomen <gilom005@umn.edu>

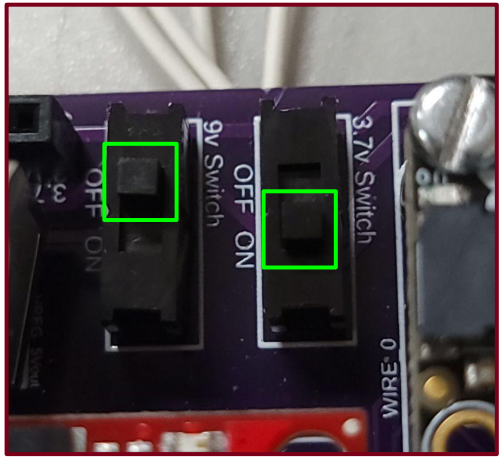
Power options - 3.7V (Recommended)

The best way to power a PTERODACTYL is by using a blue 3.7V battery as shown in the image. Simply plug the battery into the JST connector attached to the PTERODACTYL.





Completed assembly

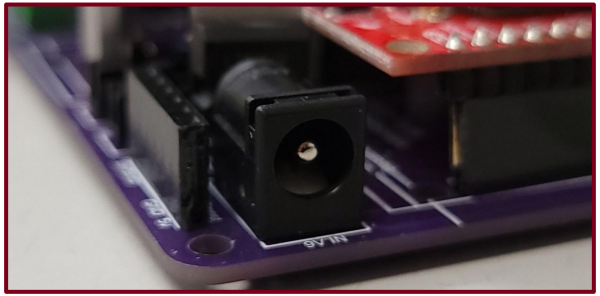


To use a 3.7V battery, make sure the two switches (located right next to the “9V IN” port) are in the states shown above. 3.7v must be “on” and 9v must be “off”.

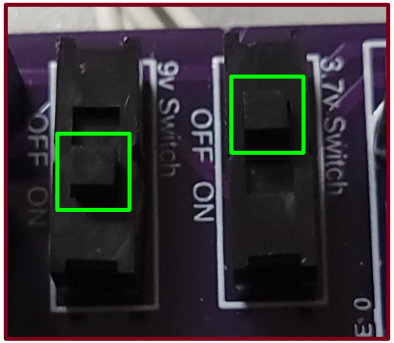
Power options - 9V Battery

Note: If you use a 9V battery for stratospheric balloon flights, be sure it is an Energizer Ultimate Lithium battery, not a Duracell battery.

To connect a 9V battery, a 9V battery jack must be used. The round connector side of the jack can connect into the port labelled "9V IN" next to the M9N GPS.



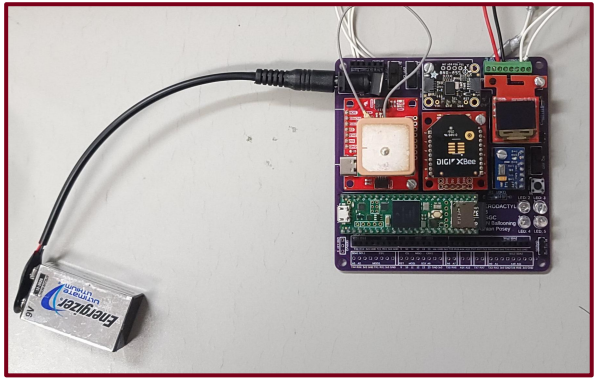
"9V IN" port close up



To use a 9V battery, make sure the two switches (located right next to the "9V IN" port) are in the states shown above. 9v must be "on" and 3.7v must be "off".



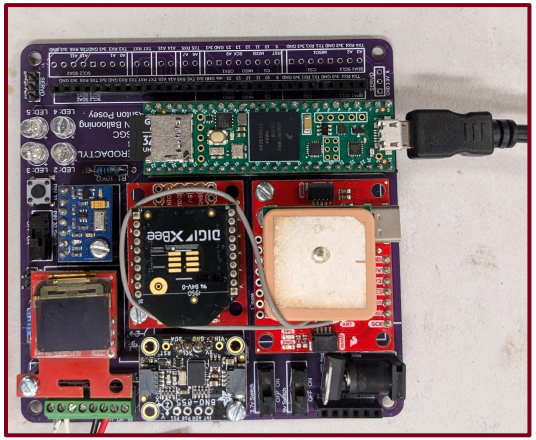
9V Battery jack



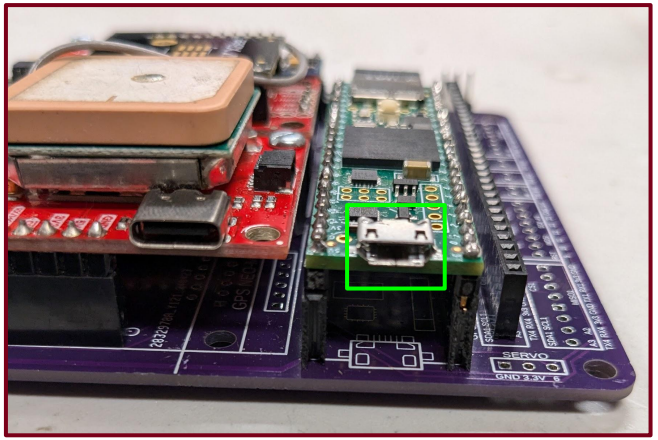
Completed assembly

Power options - Micro USB (5V)

To power the PTERODACTYL using a micro USB connector, simply plug the micro USB cable into the Teensy 4.1 board. This will also allow code to be uploaded to the Teensy.
No special switch configurations are needed for this setup.
Make sure the USB supply being used can provide at least 5V 2A DC current.

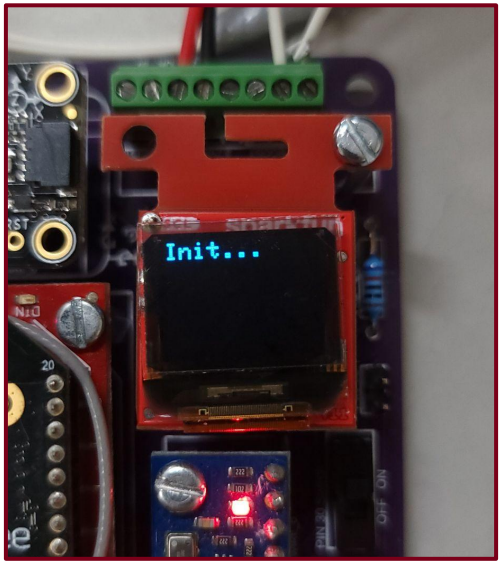


Connected assembly

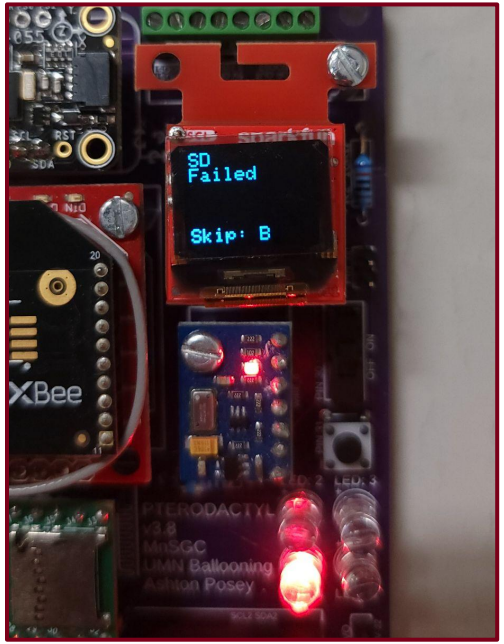


Micro USB port close up

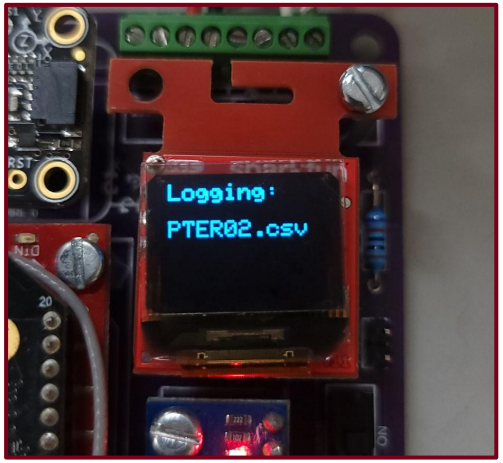
Turn-on screens



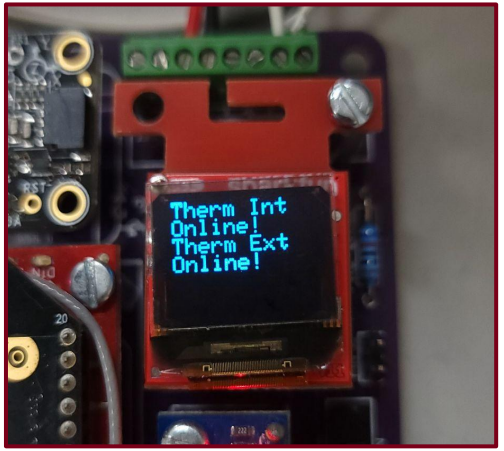
On startup, the PTERODACTYL will initialize



If there is no SD card or there is an SD card error, it will be shown on the OLED and the red LED will light up.



Name of the file the data will be logged to.



Thermistor working message.

New setup

Setup is done with the UI on the OLED screen, and the input switch and button below the OLED screen

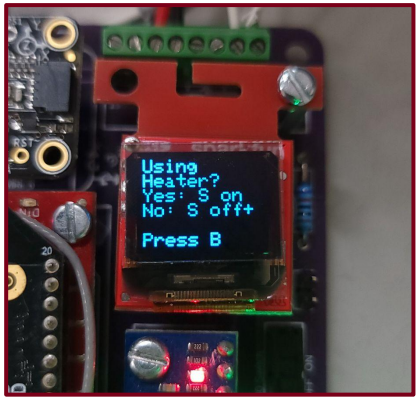


On the first startup with an empty SD card, select “yes”. This creates a new setup file on the SD card. After this, you can select no or wait 10 seconds and it will check for an old setup file



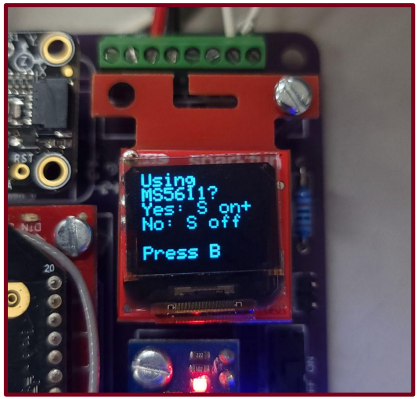
This switch (S) and button (B) serve as the input for this setup

Inputting specifications

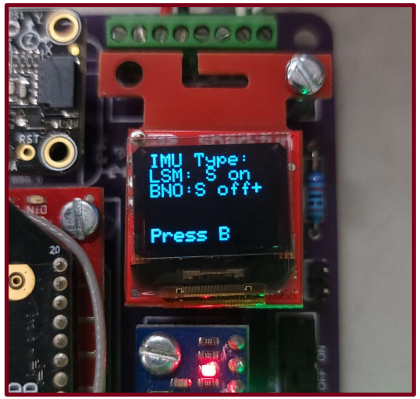


If a heater is connected to the PTERODACTYL, select yes. Normally, this is set to “No”.

External heaters connected to external batteries that are not connected electrically to the PTER do not count



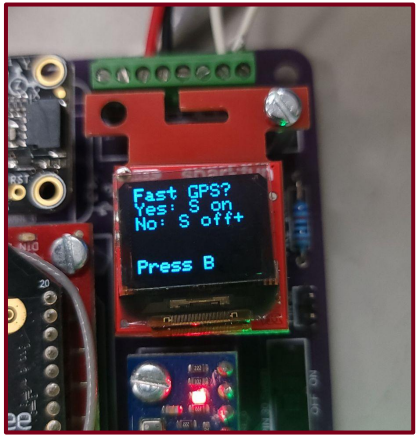
The pressure sensor on the PTERODACTYL is an MS5611, so the correct input is “Yes”.



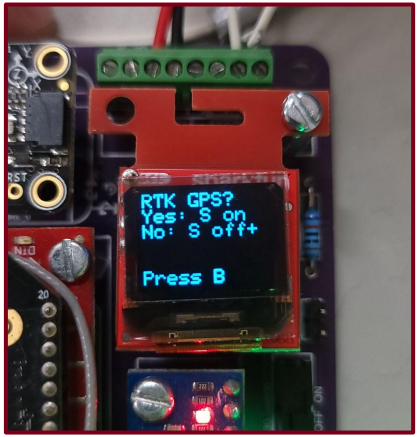
The IMU on the PTER V3.8 is a BNO-055. This is best, as LSM IMUs are no longer in production

Inputting GPS specifications

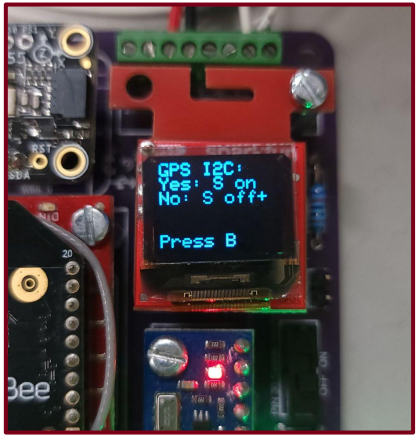
Note: On this variant of the PTERODACTYL pcb the GPS module is far enough away from the Teensy so that it (the gps module) does not need to be mounted “off-board” on a cable. On the other hand, when you start up PTERODACTYL (outside) we suggest you leave the payload box open so the gps module has the best-possible view of the sky. Don’t seal the payload box until after the gps has gotten a lock.



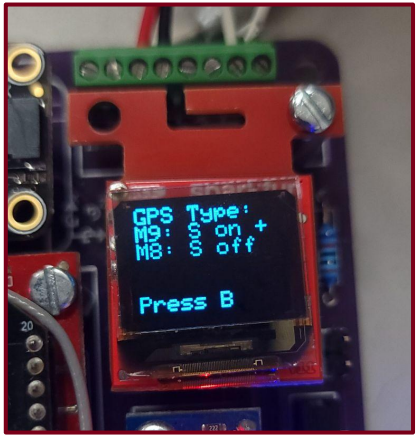
The M9N GPS on the PTER V3.8 is not “Fast”, so the input here is “No”



The M9N GPS is not RTK, so the input here is “No”.

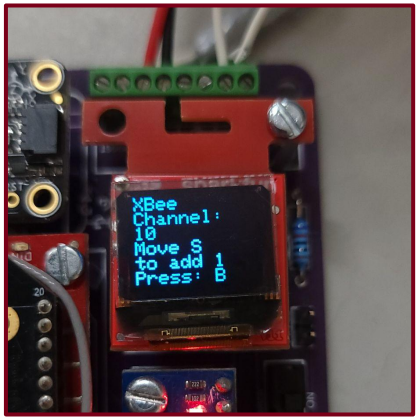


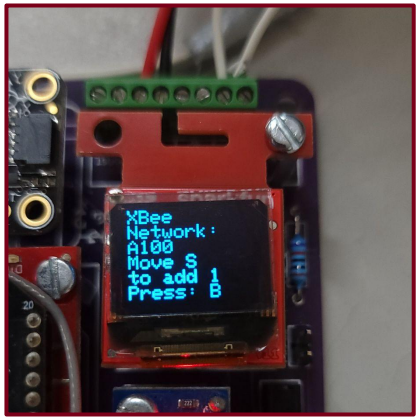
The standard mode of connection is Serial, so the standard input is “No”. however, there is an I2C connection on the board, so if the GPS is adjusted and connected via the I2C port, the input is “Yes”

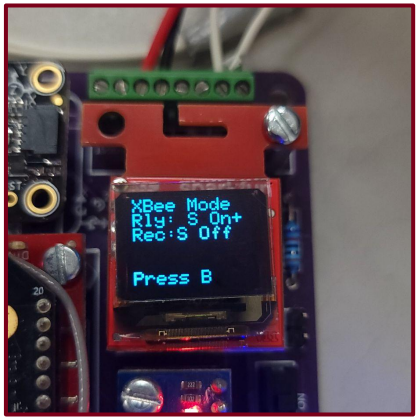


The GPS on the PTER V3.8 is a M9N GPS

Inputting XBee specifications

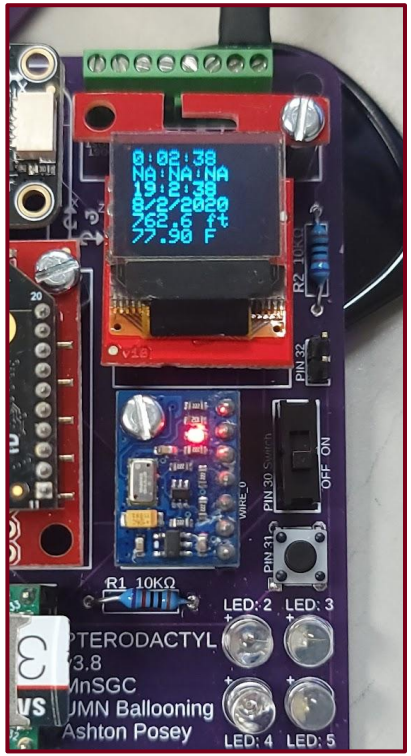






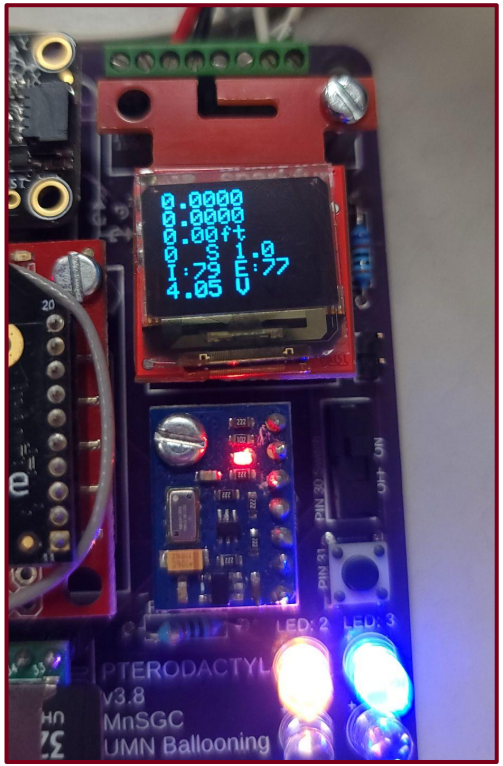
This setup screen help you configure the XBee radio for communicating with other modules. If not being used, you can click the button as many times as needed to skip through the screens.

Data feedback screen(s)



When the switch is off the screens rotate what info they show

This screen turns off when a drop in pressure is sensed, to reduce power consumption during a flight



When the switch is on, the screen stays on whatever screen is shown whenever the switch was flipped on